PRODUCT USER MANUAL

Atlantic - European North West Shelf - Ocean Wave Analysis and Forecast Product

NWSHELF_ANALYSISFORECAST_WAV_004_014

Issue: 3.0

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2020/08	All	Merged PUM_004_013 and 004_014. Updated for Dec 2020 EIS.	M. Tonani	
1.1	2021/01	All	Updated for May	I. Ascione	
1.2	2021/08	II.3 and II.4	Details on vertical grid and vertical interpolation	M. Tonani and I. Ascione	
1.3	2022/01	II.2 and II.4	Upgrade to GLO-HR boundary conditions and new MDT	A. Saulter and I. Ascione	C. Derval
2.0	2023/06/16	All	**[EIS September 2023]** First release of the NWS wave analysis and forecast product, as an extraction of the IBI operational system.	A. Amo, C. Toledano, L. Castrillo, R. Aznar, L. Aouf	M.G. Sotillo, S. Cailleau
2.1	2024/05/31	All	**[EIS November 2024]** Model system update: Resolution increased from 1/20º to 1/36º. Assimilation system upgrade: inclusion of SWOT-NADIR and HY2B altimetric data.	L. Castrillo, B. Gómez, R. Aznar, L. Aouf	M.G. Sotillo, S. Cailleau
3.0	2025/05	All	**[EIS November 2025]** Re-introduction of the NWS wave analysis and forecasting product by the UK Met Office	C.Stokes, J.Tinker, T.Brüning	Copernicus Marine Product Management

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GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
NWS	North-West Shelf
TAC	Thematic Assembly Center
NRT	Near Real Time
PUM	Product User Manual
QUID	Quality Information Document
CMEMS	Copernicus Marine Environment Monitoring Service
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
MDS	Marine Data Store
FORECAST	An assessment of the future state of the ocean variables made using numerical models.
HINDCAST	An assessment of the past state of the ocean variables made using numerical models with or without data assimilation. Simulations and analyses are hindcasts.
SIMULATION	A hindcast done without data assimilation.
ANALYSIS	A hindcast adjusted with data assimilation
SSS	Sea Surface Salinity
SSC	Sea Surface Current
SSH	Sea Surface Height
SST	Sea Surface Temperature
SMC	Spherical Multiple-Cell variable resolution wave model grid
Zonal Velocity	West to East component of the horizontal velocity vector
Meridional Velocity	South to North component of the horizontal velocity vector

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is the user manual for the North West-European Shelf analysis and forecast product:

NWSHELF_ANALYSISFORECAST_WAV_004_014

This product is generated by a coupled physical-wave forecasting system. The corresponding physical product is **NWSHELF_ANALYSISFORECAST_PHY_004_013**, which is described here.

Every day:

- the time series (analysis for 2 days before present and 7 days forecast) is updated at the end of the operational production process
- old forecasts are deleted from the MDS every day

The product is published on and available through the MDS after automatic and human quality controls. Downloaded files are in NetCDF format and follow CF-1.7 convention.

The wave analysis and forecasting system is described in the Quality Information Document (QUID):

https://documentation.marine.copernicus.eu/QUID/CMEMS-NWS-QUID-004-014.pdf

More detailed information can be obtained from the Copernicus Marine Service Team.

b) History of changes

Date	Description of changes and impacted product
July 2020	Correction of 30 min shift in time record of hourly dataset.
December 2020	Removal of TMB dataset Addition of SST, SSS, SSC, MLD, BED hourly datasets Addition of SSC and SSH 15min datasets time units in seconds since 1970-01-01T00:00:00Z
EIS November 2023	First release of the NWS wave analysis and forecast product, as an extraction of the IBI operational system.
EIS November 2024	Model system update: Resolution increased from 1/20º to 1/36º. Assimilation system upgrade: inclusion of SWOT-NADIR and HY2B altimetric data.
EIS November 2025	Re-introduction of the NWS wave analysis and forecasting product by the UK Met Office including a 7-day forecast

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

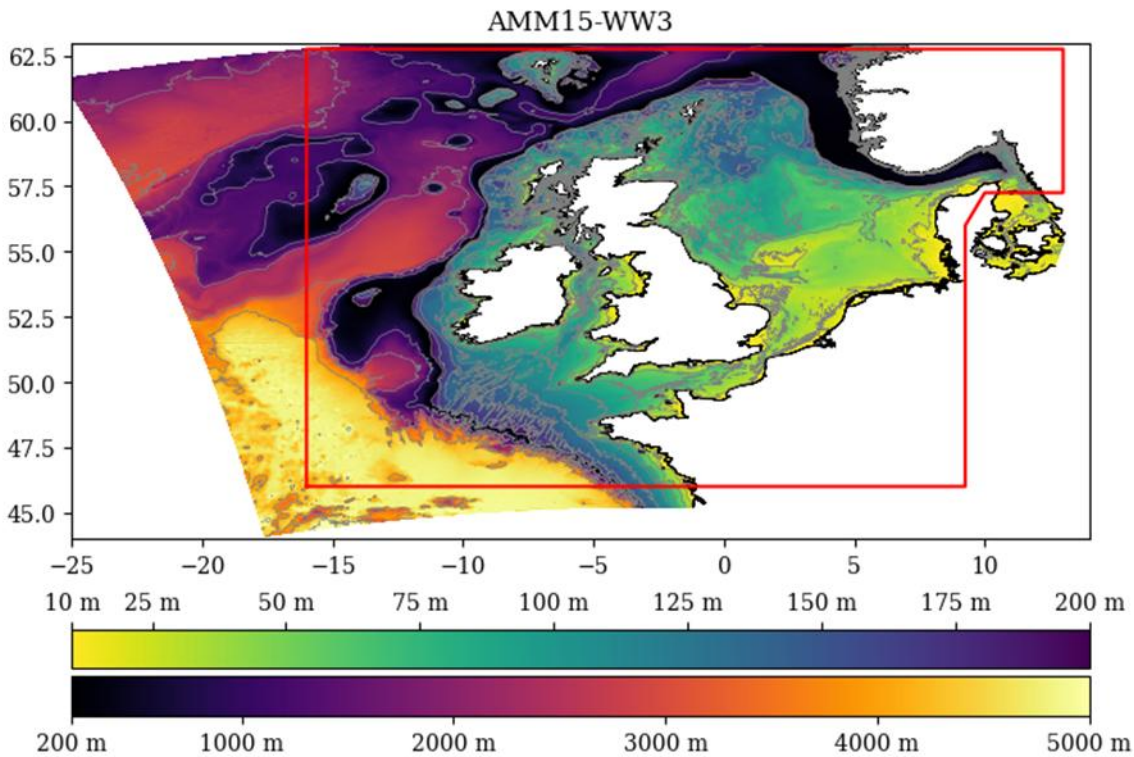
a) General Information about products

Product Lines	NORTHWESTSHELF_ANALYSIS_FORECAST_WAV_004_014	
Geographical coverage	16°W -> 13°E; 46°N -> 62.75°N. The grid points in the Baltic Sea resp. the Kattegat South of 57.25°N are masked.	
Variables	Spectral significant wave height (Hm0) expected maximum wave height expected maximum wave height from crest Spectral moments (-1,0) wave period (Tm-10) Spectral moments (0,2) wave period (Tm02) Wave period at spectral peak / peak period (Tp) Mean wave direction from (Mdir) Wave principal direction at spectral peak Stokes drift U Stokes drift V Spectral significant wind wave height Spectral moments (0,1) wind wave period Mean wind wave direction from Spectral significant primary swell wave height Spectral moments (0,1) primary swell wave period Mean primary swell wave direction from Spectral significant secondary swell wave height Spectral moments (0,1) secondary swell wave period Mean secondary swell wave direction from	
Product Type	Near Real Time - Analysis	Near Real Time - Forecast
Available time series	2 years before present	7 days forecast
Temporal resolution	Hourly instantaneous	Hourly instantaneous
Update Frequency	Daily	Daily
Target delivery time	daily at 12 pm (noon) UTC	
Horizontal resolution	1.9± 0.4 km longitude x 1.5km latitude 1/33° longitude x 1/74° latitude	
Number of vertical levels	Wave data is only delivered at the surface	
Format	NetCDF4 – CF1.7	
Delivery mechanism	Copernicus Marine Toolbox	

b) Production System Description

The operational North-West-European Shelf analysis and forecast products are generated by a coupled physical-wave forecasting system at 1/33 degree (longitude) and 1/74° (latitude). It is providing 7 days of 3D (PHY_004_013) or surface (WAV_004_014) model predictions, updated daily. This product (the wave product) includes hourly instantaneous files. Rolling archive of 2 years is available.

NORTH WEST EUROPEAN SHELF (16°W - 13°E; 46°N - 63°N)
1/33° longitude x 1/74° latitude (958 x 1240 grid points)

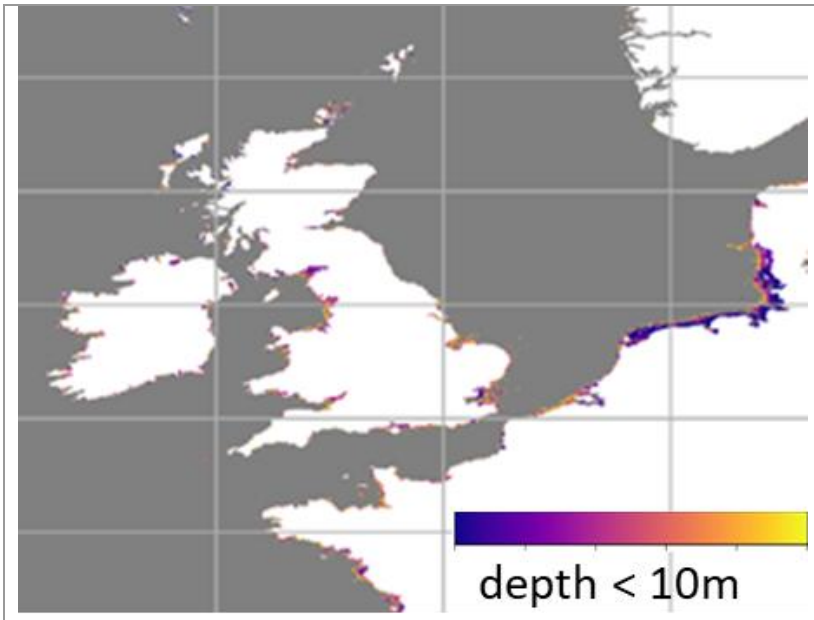


NWS bathymetry (m) showing the AMM15 model domain, with the red line showing the product domain.

The domain extends beyond the continental shelf in order to place the model's boundary region in the deep waters of the adjacent North-East Atlantic, but the focus region for the model comprises open waters of the shelf seas, i.e. (using UK terminology) the North Sea, Irish Sea, English Channel, Celtic Sea and Bay of Biscay.

The model is a nested regional ocean model configuration, coupled with a physical model. PHY and WAV products are delivered on the same horizontal grid.

A minimum depth of 10m is imposed in the ocean model for considering the sea level tidal excursion. Products in areas where the bathymetry is less than 10m could be affected by this modelling choice.



1) What is done:

We literally take every bathymetric grid point < 10 m in the domain and reassign it with a depth of 10 m, shown as coloured points in the figure above. The adjusted depths are used in the model computations. These grid points are interpolated back to their original depth before delivering the file to the CMEMS catalogue. Clearly areas such as the Wadden Sea are particularly affected (see below reference for a case study in the Wadden sea).

2) Why it is done:

This is because version 3.6 of NEMO does not have wetting and drying capability. If the bathymetry were not modified, it would mean that with the large tides that exist in this domain the model would not be able to handle the case when the tide goes out.

The minimum depth prevents such a situation from occurring and keeps the model as a whole stable.

3) What is the likely effect.

The propagation of waves, tides, and surges are inherently related to the water depth. The speed is proportional to $\sqrt{gH}$ where H is the water depth and thus as the model water depth is deeper than reality the propagation will be incorrect. In the WAV product, wave propagation is also affected by interaction with tidal currents. As such, wave propagation and wave-tide interaction will be incorrect at depths close to or less than 10 m. In cases where the seabed is exposed then this process is completely absent from the model by design (there will be water in the model where in reality there is none). It is thus not recommended to the user to directly use the model fields in regions where there are extensive areas of bathymetry less than 10 m in depth. If a model is to be nested within this product, then it is recommended that the boundaries of the nested domain are sufficiently far from such areas to reduce their impact.

4) Wetting and Drying is implemented in NEMO 4.

This specific problem is addressed with the wetting and drying module in NEMO 4 which will be part of future development of this operational Copernicus product.

The interested reader is directed to the recent paper on its development:

https://doi.org/10.1016/j.ocemod.2020.101708

Enda O'Dea, Michael J. Bell, Andrew Coward, Jason Holt, Implementation and assessment of a flux limiter based wetting and drying scheme in NEMO, Ocean Modelling, Volume 155,2020.

Horizontal grid	1.9± 0.4 km longitude x 1.5km latitude The longitude grid resolution varies with latitude. It is coarser at lower latitude (lat=46°N ->2.3 km), and finer at higher latitude (lat=63°N ->1.5km). The latitudinal grid resolution is constant in the whole model domain and is 1.5km).

	It should be noted that whilst the North-West Shelf physical product is provided on a regular grid at 1/33° longitude x 1/74° latitude, this is derived from a model with a native grid using a resolution of 1/60°. An Euler rotation is applied to the spherical - polar grid such that a pseudo equator passes through the domain, resulting in a grid with a regular resolution of approximately 1.8 km both zonally and meridionally (Guihou et al. 2017). The wave model native horizontal grid is an SMC grid (Li and Saulter, 2014) with a variable resolution of 3-1.5 km, designed for improved resolution of islands and coastlines.
Vertical grid	Wave models describe ocean surface conditions only, so do not require a vertical coordinate system. Furthermore, the majority of effects on wave propagation and dissipation occur in waters of intermediate or shallow depth as defined relative to the wave frequency (inverse of wave period): $d \leq \frac{g}{4\pi f^2}$; where d is water depth, g acceleration due to gravity and f is wave frequency. Thus, bathymetry is only a controlling mechanism on the wave field for depths below approximately 490 m, based on a minimum frequency in the model of approximately 0.04 Hz (period 25 seconds).
Spectral grid	WWIII-AMM15 wave spectra are defined on a two-dimensional frequency-direction grid using 36 (10°) direction cells and 30 geometrically spaced frequency cells. The frequency spacing is used to enable an efficient application of the model's Discrete Interaction Approximation parameterisation of nonlinear wave-wave interactions (Hasselmann et al., 1985) and is based on a multiplication factor of 1.1. The minimum frequency for the configuration is set at 0.04118 Hz, such that the maximum frequency bin is at 0.6532 Hz and the prognostic part of the wave model covers periods from approximately 25 to 1.5 seconds. In order to include the important contribution of higher frequency waves to wave growth/dissipation processes and the output wave characteristics a parametric tail is fitted for frequencies above the spectral maximum (e.g. WAMDIG, 1988), in this case using an f^{-5} relationship (Tolman, 2014).
System Name	AMM15
Model version	WAVEWATCH III v 7.1 coupled with physical model NEMO3.6 (Madec et al., 2017)
Coupling	2-way on-line ocean-wave coupling via OASIS3-MCT. • Modification of water-side surface stress on the ocean by wave growth and dissipation • Stokes-Coriolis force in the momentum equation • Wave-height-dependent ocean surface roughness • The wave model is forced by the ocean surface currents. Coupling frequency: hourly.
Bathymetry	EMODnet 2015. In order to match the grid set-up in the hydrodynamic model, but improve the estimate of wave energy transmission in the vicinity of sub grid scale topographic features such as small islands and coastal

	headlands, a number of sub grid blocking cells are defined for the wave model following the method described by Chawla and Tolman (2008).
Initial conditions	At each forecast cycle, the model is restarted from the wave conditions at the relevant time from the previous cycle. The initial conditions are constrained over successive cycles by including a 48 hour hindcast run of the model prior to each forecast. The role of the hindcast is to apply analysed wind and surface current data to the wave model, so that the model is forced by the best available descriptions of atmosphere and ocean.
Atmospheric forcings	Met Office Global Unified Model (MetUM) wind at 10 km horizontal resolution
Lateral boundaries	The WWIII-NWS model is nested one-way with lateral boundary conditions supplied from the Met Office global wave model (Saulter et al., 2016). The wave boundary data are supplied as wave spectra which are interpolated from the global model's 25 km cell resolution onto the NWS regional model SMC grid resolution (3 or 1.5 km, depending on water depth).

c) Dataset details

NORTHWESTSHELF_ANALYSIS_FORECAST_WAV_004_014	
Hourly Instantaneous (**PT1H-i**) fields of various wave parameters describing the sea surface (2D)	
Datasets	Variables name in the NetCDF file [Unit] Long_name CF Standard_name
	VHM0 [m] Spectral significant wave height (Hm0) sea_surface_wave_significant_height
	VMXL [m] expected maximum wave height from crest (linear, 1st order) sea_surface_wave_maximum_crest_height
	VCMX [m] expected maximum wave height (linear, 1st order) sea_surface_wave_maximum_height
	VTM10 [s] Spectral moments (-1,0) wave period (Tm-10) sea_surface_wave_mean_period_from_variance_spectral_density_inverse_frequency_moment
	VTM02 [s] Spectral moments (0,2) wave period (Tm02) sea_surface_wave_mean_period_from_variance_spectral_density_second_frequency_moment
	VTPK [s] Wave period at spectral peak / peak period (Tp) sea_surface_wave_period_at_variance_spectral_density_maximum
	VMDR [degree] Mean wave direction from (Mdir)

cmems_mod_nws_wav_anfc_1.5km_PT1H-i	sea_surface_wave_from_direction
	VPED [degree] Wave principal direction at spectral peak sea_surface_wave_from_direction_at_spectral_density_maximum
	VSDX [m s-1] Stokes drift U sea_surface_wave_stokes_drift_x_velocity
	VSDY [m s-1] Stokes drift V sea_surface_wave_stokes_drift_y_velocity
	VHM0_WW [m] Spectral significant wind wave height sea_surface_wind_wave_significant_height
	VTM01_WW [s] Spectral moments (0,1) wind wave period sea_surface_wind_wave_mean_period
	VMDR_WW [degree] Mean wind wave direction from sea_surface_wind_wave_from_direction
	VHM0_SW1 [m] Spectral significant primary swell wave height sea_surface_primary_swell_wave_significant_height
	VTM01_SW1 [s] Spectral moments (0,1) primary swell wave period sea_surface_primary_swell_wave_mean_period
	VMDR_SW1 [degree] Mean primary swell wave direction from sea_surface_primary_swell_wave_from_direction
	VHM0_SW2 [m] Spectral significant secondary swell wave height sea_surface_secondary_swell_wave_significant_height
	VTM01_SW2 [s] Spectral moments (0,1) secondary swell wave period sea_surface_secondary_swell_wave_mean_period
	VMDR_SW2 [degree] Mean secondary swell wave direction from sea_surface_secondary_swell_wave_from_direction
cmems_mod_nws_wav_anfc_1.5km_static	**mask** Land-sea mask sea_binary_mask
	depth [m] depth depth
	deptho [m] Bathymetry sea_floor_depth_below_geoid

The model **bathymetry** for this product is the same as for the product NWSHELF_ANALYSISFORECAST_PHY_004_013 as both products are obtained from a coupled physical-wave model system.

d) Additional information on some parameters

VSDX, VSDY [m s-1]	The total Lagrangian velocity (Surface and Merged Ocean Currents, SMOC, in GLOBAL_ANALYSIS_FORECAST_001_024 product) can be obtained combining the eulerian ocean horizontal velocity (uo, vo) from product NWSHELF_ANALYSISFORECAST_PHY_004_013 and the Stokes drift (VSDX, VSDY) from this product. The Stokes drift provided by this product is the wave-induced drift at the surface $\boldsymbol{u_s^0}$(t,x,y,0). The 3D Stokes' drift $\boldsymbol{u_s}$(t,x,y,z) can be estimated using the surface wave-induced drift $\boldsymbol{u_s^0}$, the significant wave height H_s $(VHMO)$ and the mean wave period T_{01} as proposed by Breivik et al. 2016: $$\boldsymbol{u_s} = \boldsymbol{u_s^0}\,[\boldsymbol{e^{2k_e z}} - \boldsymbol{\beta}\sqrt{-\boldsymbol{2k_e \pi z}}\,\boldsymbol{erfc}(\sqrt{\boldsymbol{2k_e z}})]$$ where $\beta = 1$; $k_e = \frac{8\lvert \boldsymbol{u_s^0} \rvert T_{01}}{\gamma \pi H_s^2}$ with $\gamma = 5.97$ and z the depth. The 3D Stokes' drift $\boldsymbol{u_s}$ and the Eulerian ocean currents $\boldsymbol{u}$ can be linearly combined to obtain the total Lagrangian velocity $\boldsymbol{u_L} = \boldsymbol{u} + \boldsymbol{u_s}$. The surface ocean horizontal velocity is referred to a layer with 1m thickness. Therefore, the Stokes drift at 0.5 m depth should be used for computing the total Lagrangian velocity for the first ocean level.

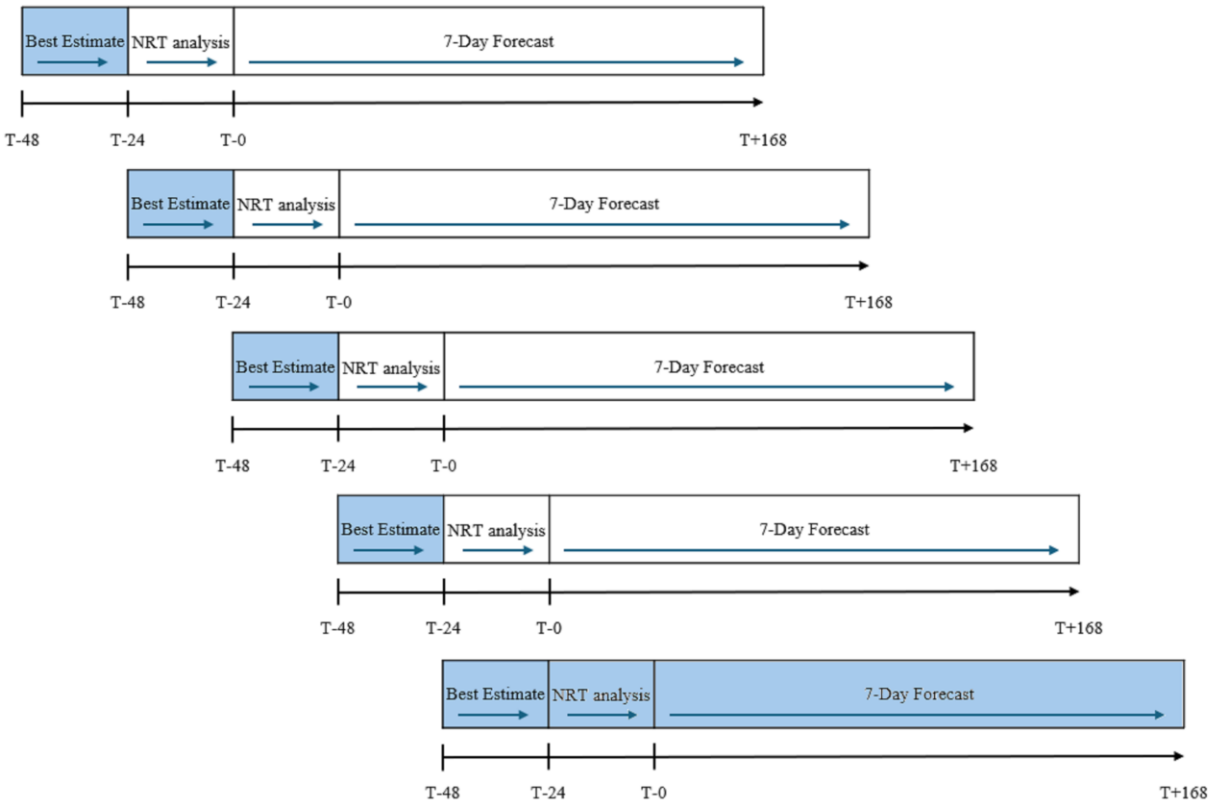
e) Processing information

Product disseminating schedule :

Type of product (forecast/ hindcast)	Forecast run (base time)	Product temporal coverage (hours)	Product disseminating scheduled time	Dissemination day
Forecast	00 UTC	0 to 168	until 12 UTC	Every day
NRT - Analysis	00 UTC	-24 to 0	until 12 UTC	Every day
Best estimate - Analysis	00 UTC	-48 to -24	until 12 UTC	Every day

f) Catalogue updating

The time coverage of the catalogue data spans from two years before present to present plus 7 days of forecast.



Each day the [T-48h, T-24h] and [T-24h, T+00h] analyses are provided along with 7 days forecast for this product (with the first day of the forecast being for the day of production).
The delivery usually happens between 6:40 AM and 10:00 AM, a couple of hours before the target delivery time. The previous day 'NRT analysis' and '7-day of forecast' are deleted at the beginning of this process. The catalogue data from 2-year before present to present minus 24 hours is taken from T-48h to T-24h of each run (Best Estimate - Analysis).

g) Time averaging

Dataset frequency	Time averaging	Forecast lead time
hourly instantaneous values; time steps are in UTC	No temporal averaging or interpolation performed.	7-day (168-hr)

h) Product use and impact

Specialised marine waves analysis and forecast product for the Northwest Shelf region. Uses include marine navigation, emergency response (search and rescue/pollution), coastal flood mitigation, health and ecosystems monitoring, energy and net zero. Applications are developed for government policy, industrial decision making and academic research. The NWS products provide high resolution data and forecasts for ocean state in the tidally dominated waters of the Northwest Shelf and are designed to provide state of the art information specific to this region. The wave forecast data are used routinely by coastal flood forecasters from the Met Office, UK Environment Agency, Scottish Environment Protection Agency, Natural Resources Wales, and Department for Infrastructure Rivers Directorate Ireland. These data provide forecasters with information on swell and windsea conditions to inform warnings around coastal wave overtopping, flooding, and hazardous waves at the coast.

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

4.1) FILE SIZE

NORTHWESTSHELF_ANALYSIS_FORECAST_WAV_004_014		
DATASET NAME	**NAME OF FILE**	**DIMENSION [MB]**
cmems_mod_nws_wav_anfc_1.5km_PT1H-i	metoffice_wave_amm15_NWS_WAV_**b**{{yyyymmdd}}_**hi**{{yyyymmdd}}.nc	261
cmems_mod_nws_wav_anfc_1.5km_static	NWS-MFC_004_014_static.nc	10

b = the bulletin date (date when the forecast was produced)
hi = hourly instantaneous data (frequency of the data within the file)

4.2) STRUCTURE OF FILES

An example of the header of output NetCDF files is inserted in **annex**.

5) REFERENCES

Quality Information Document
https://documentation.marine.copernicus.eu/QUID/CMEMS-NWS-QUID-004-014.pdf

Breivik Ø., Bidlot J.R., Janssen P.A., (2016). A Stokes drift approximation based on the Phillips spectrum. Ocean Model.,100,49-56.

Chawla A., Tolman H.L., (2008). Obstruction grids for spectral wave models, Ocean Modelling,22,12-25.

Guihou K., Polton J., Harle J., Wakelin S., O'Dea E., Holt J., (2017). Kilometric scale modelling of the North West European shelf seas: exploring the spatial and temporal variability of internal tides, J. Geophys. Res.-Oceans, 122. https://doi.org/10.1002/2017JC012960

Hasselmann S, Hasselmann K, Allender J.H., Barnett, T.P., (1985). Computations and parametrizations of the nonlinear energy transfer in a gravity-wave spectrum. Part 2: Parametrizations of the nonlinear energy transfer for application in wave models, J. Phys. Oceanogr.,15,1378-1391.

Li, J.G. and Saulter, A., (2014). Unified global and regional wave model on a multi-resolution grid. Ocean Dynamics, 64, pp.1657-1670.

Saulter, A., Bunney, C. and Li, J., (2016). Application of a refined grid global model for operational wave forecasting. Met Office: Exeter, UK.

Tolman H. L., (2014). User manual and system documentation of WAVEWATCH III version 4.18. NOAA / NWS / NCEP / MMAB Technical Note 316, 194 pp.+ Appendices (pdf).

WAMDIG, (1988). The WAM model - A third generation ocean wave prediction model. Journal of Physical Oceanography, 18, 1775-1810.

6) ANNEX

a) Structure and semantic of NetCDF maps files

```
dimensions:
        longitude = 958 ;
        latitude = 1240 ;
        time = UNLIMITED ; // (24 currently)
variables:
        float longitude(longitude) ;
                longitude:units = "degree_east" ;
                longitude:valid_min = -180.f ;
                longitude:axis = "X" ;
                longitude:long_name = "longitude" ;
                longitude:standard_name = "longitude" ;
                longitude:valid_max = 180.f ;
        float latitude(latitude) ;
                latitude:units = "degree_north" ;
                latitude:valid_min = -90.f ;
                latitude:valid_max = 90.f ;
                latitude:axis = "Y" ;
                latitude:long_name = "latitude" ;
                latitude:standard_name = "latitude" ;
        double time(time) ;
                time:long_name = "time" ;
                time:standard_name = "time" ;
                time:axis = "T" ;
                time:calendar = "gregorian" ;
                time:units = "seconds since 1970-01-01T00:00:00Z" ;
        int forecast_period(time) ;
                forecast_period:long_name = "forecast period" ;
                forecast_period:standard_name = "forecast_period" ;
                forecast_period:units = "s" ;
        double forecast_reference_time ;
                forecast_reference_time:long_name = "forecast reference time" ;
                forecast_reference_time:standard_name = "forecast_reference_time" ;
                forecast_reference_time:calendar = "gregorian" ;
                forecast_reference_time:units = "seconds since 1970-01-01T00:00:00Z" ;
        int crs ;
                crs:grid_mapping_name = "latitude_longitude" ;
                crs:longitude_of_prime_meridian = 0.f ;
                crs:semi_major_axis = 6378137.f ;
                crs:inverse_flattening = 298.2572f ;
        short VHM0(time, latitude, longitude) ;
                VHM0:long_name = "Spectral significant wave height (Hm0)" ;
                VHM0:standard_name = "sea_surface_wave_significant_height" ;
                VHM0:units = "m" ;
                VHM0:scale_factor = 0.002f ;
                VHM0:add_offset = 0.f ;
                VHM0:valid_min = 0 ;
                VHM0:valid_max = 16000 ;
                VHM0:grid_mapping = "crs" ;
                VHM0:coordinates = "forecast_period forecast_reference_time" ;
```

```
        VHM0:cell_methods = "time:point" ;
        VHM0:_FillValue = -32767s ;
short VTM02(time, latitude, longitude) ;
        VTM02:long_name = "Spectral moments (0,2) wave period Tm02" ;
        VTM02:standard_name =
        "sea_surface_wave_mean_period_from_variance_spectral_density_second_frequency_
        moment" ;
        VTM02:units = "s" ;
        VTM02:scale_factor = 0.01f ;
        VTM02:add_offset = 0.f ;
        VTM02:valid_min = 0 ;
        VTM02:valid_max = 3000 ;
        VTM02:grid_mapping = "crs" ;
        VTM02:coordinates = "forecast_period forecast_reference_time" ;
        VTM02:cell_methods = "time:point" ;
        VTM02:_FillValue = -32767s ;
short VTM10(time, latitude, longitude) ;
        VTM10:long_name = "Spectral moments (-1,0) wave period (Tm-10)" ;
        VTM10:standard_name =
        "sea_surface_wave_mean_period_from_variance_spectral_density_inverse_frequency_
        moment" ;
        VTM10:units = "s" ;
        VTM10:scale_factor = 0.01f ;
        VTM10:add_offset = 0.f ;
        VTM10:valid_min = 0 ;
        VTM10:valid_max = 3000 ;
        VTM10:grid_mapping = "crs" ;
        VTM10:coordinates = "forecast_period forecast_reference_time" ;
        VTM10:cell_methods = "time:point" ;
        VTM10:_FillValue = -32767s ;
short VMDR(time, latitude, longitude) ;
        VMDR:long_name = "Mean wave direction from (Mdir)" ;
        VMDR:standard_name = "sea_surface_wave_from_direction" ;
        VMDR:units = "degree" ;
        VMDR:scale_factor = 0.1f ;
        VMDR:add_offset = 0.f ;
        VMDR:valid_min = 0 ;
        VMDR:valid_max = 3600 ;
        VMDR:grid_mapping = "crs" ;
        VMDR:coordinates = "forecast_period forecast_reference_time" ;
        VMDR:cell_methods = "time:point" ;
        VMDR:direction_reference = "True North" ;
        VMDR:_FillValue = -32767s ;
short VPED(time, latitude, longitude) ;
        VPED:long_name = "Wave principal direction at spectral peak" ;
        VPED:standard_name =
        "sea_surface_wave_from_direction_at_variance_spectral_density_maximum" ;
        VPED:units = "degree" ;
        VPED:scale_factor = 0.1f ;
        VPED:add_offset = 0.f ;
        VPED:valid_min = 0 ;
        VPED:valid_max = 3600 ;
        VPED:grid_mapping = "crs" ;
        VPED:coordinates = "forecast_period forecast_reference_time" ;
```

```
        VPED:cell_methods = "time:point" ;
        VPED:direction_reference = "True North" ;
        VPED:_FillValue = -32767s ;
short VTPK(time, latitude, longitude) ;
        VTPK:long_name = "Wave period at spectral peak / peak period (Tp)" ;
        VTPK:standard_name =
        "sea_surface_wave_period_at_variance_spectral_density_maximum" ;
        VTPK:units = "s" ;
        VTPK:scale_factor = 0.01f ;
        VTPK:add_offset = 0.f ;
        VTPK:valid_min = 0 ;
        VTPK:valid_max = 3000 ;
        VTPK:grid_mapping = "crs" ;
        VTPK:coordinates = "forecast_period forecast_reference_time" ;
        VTPK:cell_methods = "time:point" ;
        VTPK:_FillValue = -32767s ;
short VHM0_WW(time, latitude, longitude) ;
        VHM0_WW:long_name = "Spectral significant wind wave height" ;
        VHM0_WW:standard_name = "sea_surface_wind_wave_significant_height" ;
        VHM0_WW:units = "m" ;
        VHM0_WW:scale_factor = 0.002f ;
        VHM0_WW:add_offset = 0.f ;
        VHM0_WW:valid_min = 0 ;
        VHM0_WW:valid_max = 16000 ;
        VHM0_WW:comment = "Wind sea and swells defined using topographic partitions and
        spectral wave-age cut-off" ;
        VHM0_WW:grid_mapping = "crs" ;
        VHM0_WW:coordinates = "forecast_period forecast_reference_time" ;
        VHM0_WW:cell_methods = "time:point" ;
        VHM0_WW:_FillValue = -32767s ;
short VHM0_SW1(time, latitude, longitude) ;
        VHM0_SW1:long_name = "Spectral significant primary swell wave height" ;
        VHM0_SW1:standard_name = "sea_surface_primary_swell_wave_significant_height" ;
        VHM0_SW1:units = "m" ;
        VHM0_SW1:scale_factor = 0.002f ;
        VHM0_SW1:add_offset = 0.f ;
        VHM0_SW1:valid_min = 0 ;
        VHM0_SW1:valid_max = 16000 ;
        VHM0_SW1:comment = "Wind sea and swells defined using topographic partitions
        and spectral wave-age cut-off" ;
        VHM0_SW1:grid_mapping = "crs" ;
        VHM0_SW1:coordinates = "forecast_period forecast_reference_time" ;
        VHM0_SW1:cell_methods = "time:point" ;
        VHM0_SW1:_FillValue = -32767s ;
short VHM0_SW2(time, latitude, longitude) ;
        VHM0_SW2:long_name = "Spectral significant secondary swell wave height" ;
        VHM0_SW2:standard_name =
        "sea_surface_secondary_swell_wave_significant_height" ;
        VHM0_SW2:units = "m" ;
        VHM0_SW2:scale_factor = 0.002f ;
        VHM0_SW2:add_offset = 0.f ;
        VHM0_SW2:valid_min = 0 ;
        VHM0_SW2:valid_max = 16000 ;
```

```
        VHM0_SW2:comment = "Wind sea and swells defined using topographic partitions
        and spectral wave-age cut-off" ;
        VHM0_SW2:grid_mapping = "crs" ;
        VHM0_SW2:coordinates = "forecast_period forecast_reference_time" ;
        VHM0_SW2:cell_methods = "time:point" ;
        VHM0_SW2:_FillValue = -32767s ;
short VMDR_WW(time, latitude, longitude) ;
        VMDR_WW:long_name = "Mean wind wave direction from" ;
        VMDR_WW:standard_name = "sea_surface_wind_wave_from_direction" ;
        VMDR_WW:units = "degree" ;
        VMDR_WW:scale_factor = 0.1f ;
        VMDR_WW:add_offset = 0.f ;
        VMDR_WW:valid_min = 0 ;
        VMDR_WW:valid_max = 3600 ;
        VMDR_WW:comment = "Wind sea and swells defined using topographic partitions
        and spectral wave-age cut-off" ;
        VMDR_WW:grid_mapping = "crs" ;
        VMDR_WW:coordinates = "forecast_period forecast_reference_time" ;
        VMDR_WW:cell_methods = "time:point" ;
        VMDR_WW:direction_reference = "True North" ;
        VMDR_WW:_FillValue = -32767s ;
short VMDR_SW1(time, latitude, longitude) ;
        VMDR_SW1:long_name = "Mean primary swell wave direction from" ;
        VMDR_SW1:standard_name = "sea_surface_primary_swell_wave_from_direction" ;
        VMDR_SW1:units = "degree" ;
        VMDR_SW1:scale_factor = 0.1f ;
        VMDR_SW1:add_offset = 0.f ;
        VMDR_SW1:valid_min = 0 ;
        VMDR_SW1:valid_max = 3600 ;
        VMDR_SW1:comment = "Wind sea and swells defined using topographic partitions
        and spectral wave-age cut-off" ;
        VMDR_SW1:grid_mapping = "crs" ;
        VMDR_SW1:coordinates = "forecast_period forecast_reference_time" ;
        VMDR_SW1:cell_methods = "time:point" ;
        VMDR_SW1:direction_reference = "True North" ;
        VMDR_SW1:_FillValue = -32767s ;
short VMDR_SW2(time, latitude, longitude) ;
        VMDR_SW2:long_name = "Mean secondary swell wave direction from" ;
        VMDR_SW2:standard_name = "sea_surface_secondary_swell_wave_from_direction" ;
        VMDR_SW2:units = "degree" ;
        VMDR_SW2:scale_factor = 0.1f ;
        VMDR_SW2:add_offset = 0.f ;
        VMDR_SW2:valid_min = 0 ;
        VMDR_SW2:valid_max = 3600 ;
        VMDR_SW2:comment = "Wind sea and swells defined using topographic partitions
        and spectral wave-age cut-off" ;
        VMDR_SW2:grid_mapping = "crs" ;
        VMDR_SW2:coordinates = "forecast_period forecast_reference_time" ;
        VMDR_SW2:cell_methods = "time:point" ;
        VMDR_SW2:direction_reference = "True North" ;
        VMDR_SW2:_FillValue = -32767s ;
short VTM01_WW(time, latitude, longitude) ;
        VTM01_WW:long_name = "Spectral moments (0,1) wind wave period" ;
        VTM01_WW:standard_name = "sea_surface_wind_wave_mean_period" ;
```

```
        VTM01_WW:units = "s" ;
        VTM01_WW:scale_factor = 0.01f ;
        VTM01_WW:add_offset = 0.f ;
        VTM01_WW:valid_min = 0 ;
        VTM01_WW:valid_max = 3000 ;
        VTM01_WW:comment = "Wind sea and swells defined using topographic partitions
        and spectral wave-age cut-off" ;
        VTM01_WW:grid_mapping = "crs" ;
        VTM01_WW:coordinates = "forecast_period forecast_reference_time" ;
        VTM01_WW:cell_methods = "time:point" ;
        VTM01_WW:_FillValue = -32767s ;
short VTM01_SW1(time, latitude, longitude) ;
        VTM01_SW1:long_name = "Spectral moments (0,1) primary swell wave period" ;
        VTM01_SW1:standard_name = "sea_surface_primary_swell_wave_mean_period" ;
        VTM01_SW1:units = "s" ;
        VTM01_SW1:scale_factor = 0.01f ;
        VTM01_SW1:add_offset = 0.f ;
        VTM01_SW1:valid_min = 0 ;
        VTM01_SW1:valid_max = 3000 ;
        VTM01_SW1:comment = "Wind sea and swells defined using topographic partitions
        and spectral wave-age cut-off" ;
        VTM01_SW1:grid_mapping = "crs" ;
        VTM01_SW1:coordinates = "forecast_period forecast_reference_time" ;
        VTM01_SW1:cell_methods = "time:point" ;
        VTM01_SW1:_FillValue = -32767s ;
short VTM01_SW2(time, latitude, longitude) ;
        VTM01_SW2:long_name = "Spectral moments (0,1) secondary swell wave period" ;
        VTM01_SW2:standard_name = "sea_surface_secondary_swell_wave_mean_period" ;
        VTM01_SW2:units = "s" ;
        VTM01_SW2:scale_factor = 0.01f ;
        VTM01_SW2:add_offset = 0.f ;
        VTM01_SW2:valid_min = 0 ;
        VTM01_SW2:valid_max = 3000 ;
        VTM01_SW2:comment = "Wind sea and swells defined using topographic partitions
        and spectral wave-age cut-off" ;
        VTM01_SW2:grid_mapping = "crs" ;
        VTM01_SW2:coordinates = "forecast_period forecast_reference_time" ;
        VTM01_SW2:cell_methods = "time:point" ;
        VTM01_SW2:_FillValue = -32767s ;
short VSDX(time, latitude, longitude) ;
        VSDX:long_name = "Stokes drift U" ;
        VSDX:standard_name = "sea_surface_wave_stokes_drift_x_velocity" ;
        VSDX:units = "m s-1" ;
        VSDX:scale_factor = 0.0005f ;
        VSDX:add_offset = 0.f ;
        VSDX:valid_min = -10000 ;
        VSDX:valid_max = 10000 ;
        VSDX:comment = "uss=sqrt(uuss**2+vuss**2)" ;
        VSDX:grid_mapping = "crs" ;
        VSDX:coordinates = "forecast_period forecast_reference_time" ;
        VSDX:cell_methods = "time:point" ;
        VSDX:direction_reference = "True North" ;
        VSDX:_FillValue = -32767s ;
short VSDY(time, latitude, longitude) ;
```

```
                VSDY:long_name = "Stokes drift V" ;
                VSDY:standard_name = "sea_surface_wave_stokes_drift_y_velocity" ;
                VSDY:units = "m s-1" ;
                VSDY:scale_factor = 0.0005f ;
                VSDY:add_offset = 0.f ;
                VSDY:valid_min = -10000 ;
                VSDY:valid_max = 10000 ;
                VSDY:comment = "Frequency range   0.0412 to   0.6532 Hz" ;
                VSDY:grid_mapping = "crs" ;
                VSDY:coordinates = "forecast_period forecast_reference_time" ;
                VSDY:cell_methods = "time:point" ;
                VSDY:direction_reference = "True North" ;
                VSDY:_FillValue = -32767s ;
        short VCMX(time, latitude, longitude) ;
                VCMX:scale_factor = -3.994995e-05f ;
                VCMX:add_offset = 1.309f ;
                VCMX:cell_methods = "time:point" ;
                VCMX:coordinates = "forecast_period forecast_reference_time" ;
                VCMX:grid_mapping = "crs" ;
                VCMX:units = "m" ;
                VCMX:valid_max = 16000 ;
                VCMX:valid_min = 0 ;
                VCMX:_FillValue = -32767s ;
                VCMX:long_name = "expected maximum wave height (linear, 1st order)" ;
                VCMX:standard_name = "sea_surface_wave_maximum_height" ;
        short VMXL(time, latitude, longitude) ;
                VMXL:scale_factor = -3.994995e-05f ;
                VMXL:add_offset = 1.309f ;
                VMXL:_FillValue = -32767s ;
                VMXL:cell_methods = "time:point" ;
                VMXL:coordinates = "forecast_period forecast_reference_time" ;
                VMXL:grid_mapping = "crs" ;
                VMXL:units = "m" ;
                VMXL:valid_max = 16000 ;
                VMXL:valid_min = 0 ;
                VMXL:standard_name = "sea_surface_wave_maximum_crest_height" ;
                VMXL:long_name = "expected maximum wave height from crest (linear, 1st order)" ;
// global attributes:
                :creation_date = "2025-05-21T06:28:51Z" ;
                :Conventions = "CF-1.7" ;
                :netcdf-version-id = "netCDF-4" ;
                :title = "hourly-instantaneous wave" ;
                :institution = "UK Met Office" ;
                :contact = "servicedesk.cmems@mercator-ocean.eu" ;
                :references = "http://marine.copernicus.eu/" ;
                :source = "PS-OS 47, AMM-FOAM 1.5 km (tidal) NEMO v3.6_WAVEWATCH-III
                v7.12" ;
                :forcing_data_source = " UKMO Unified Model (MetUM) Atmospheric model
                Coupled-NWP (applied as direct fluxes); NEMO AMM15; Global WW3; UKMO
                global ORCA12 ocean model; UKMO GLOWave; CMEMS-BALTIC; climatological
                rivers" ;
                :credit = "E.U. Copernicus Marine Service Information (CMEMS)" ;
                :licence = "http://marine.copernicus.eu/services-portfolio/service-commitments-and-
                licence/" ;
```